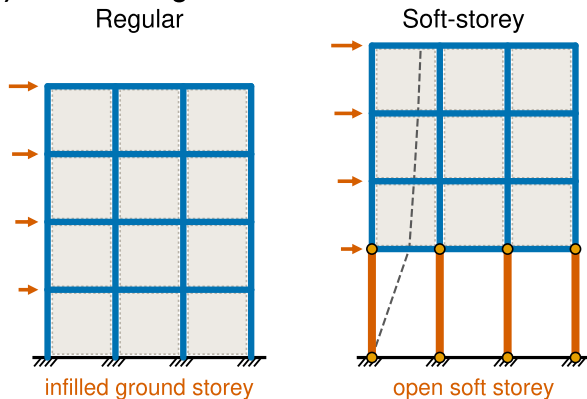
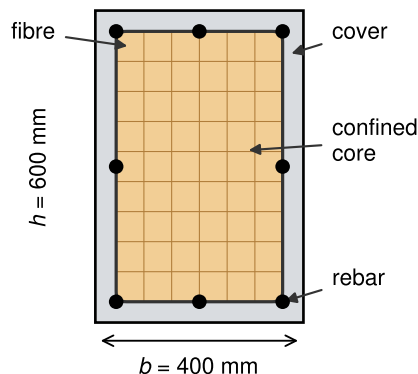


# RC frame fibre model: configurations, section discretisation and cyclic response

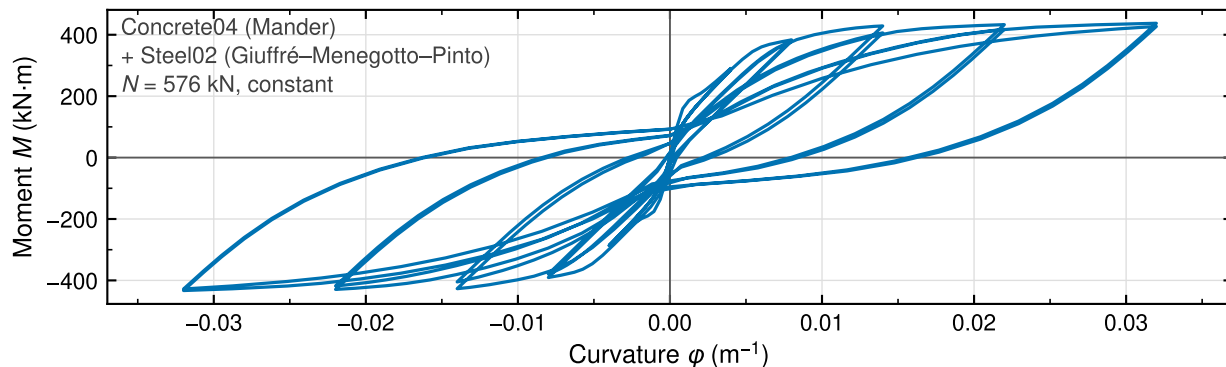
(a) Frame configurations



(b) Fibre section



(c) Cyclic moment–curvature



Depiction: (a) regular and soft-storey frame configurations, drawn to a common vertical scale; the light panels mark where masonry infill is present in the real stock and are not represented in the numerical model, which admits infill only through the ground-to-upper-storey stiffness ratio  $r_k$  of Eq. (4); (b) fibre discretisation of the 400×600 mm beam–column section into unconfined cover, confined core and longitudinal bars; (c) cyclic moment–curvature obtained by integrating the section fibres under symmetric curvature cycles, with Concrete04 (Mander/Popovics envelopes, separate confined and unconfined branches, Karsan–Jirsa unloading, linear tensile softening to zero residual stress) and Steel02 (Giuffré–Menegotto–Pinto,  $b = 0.01$ , isotropic hardening) at constant axial load  $N = 576 \text{ kN}$ . Software used for producing graph: Python 3.12.3 with Matplotlib 3.9.3 (pyplot, patches, lines, ticker, patheffects) and NumPy 2.4.6. Source: authors.